

Database Design Proposal

Submitted by: G.Pavana deepika

1. Business Scenario

TechGear is an online electronics retailer that allows customers to browse products, manage accounts, and place orders.

2. Entities and Attributes

Entity	Attributes
Customer	customer_id (PK), first_name, last_name, email (Unique), address, phone
Category	category_id (PK), category_name, description
Product	product_id (PK), product_name, price, stock_quantity, category_id (FK)
Order	order_id (PK), customer_id (FK), order_date, total_amount
Order_Item	order_item_id (PK), order_id (FK), product_id (FK), quantity, unit_price

3. ER Diagram (Mermaid Code)

Use this code in any Mermaid-compatible diagram tool (like Draw.io or Mermaid Live Editor):

```
erDiagram
    CUSTOMER ||--o{ ORDER : places
    ORDER ||--|{ ORDER_ITEM : contains
    PRODUCT ||--o{ ORDER_ITEM : included_in
    CATEGORY ||--o{ PRODUCT : contains

    CUSTOMER {
        int customer_id PK
        string email
```

```

}
ORDER {
    int order_id PK
    int customer_id FK
    datetime order_date
}
ORDER_ITEM {
    int order_item_id PK
    int order_id FK
    int product_id FK
}
PRODUCT {
    int product_id PK
    int category_id FK
}
CATEGORY {
    int category_id PK
}

```

4. SQL Schema (Implementation Code)

```

-- Create Tables
CREATE TABLE Customer (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100) UNIQUE,
    address TEXT,
    phone VARCHAR(15)
);

CREATE TABLE Category (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(50),
    description TEXT
);

CREATE TABLE Product (

```

```
product_id INT PRIMARY KEY,  
product_name VARCHAR(100),  
price DECIMAL(10, 2),  
stock_quantity INT,  
category_id INT,  
FOREIGN KEY (category_id) REFERENCES Category(category_id)  
);  
  
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATETIME,  
    total_amount DECIMAL(10, 2),  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);  
  
CREATE TABLE Order_Item (  
    order_item_id INT PRIMARY KEY,  
    order_id INT,  
    product_id INT,  
    quantity INT,  
    unit_price DECIMAL(10, 2),  
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),  
    FOREIGN KEY (product_id) REFERENCES Product(product_id)  
);
```

5. Normalization & Indexing

Normalization: Applied 1NF (atomic values), 2NF (full functional dependency), and 3NF (elimination of transitive dependencies).

Indexing Strategy:

- Unique Index on Customer(email) for efficient login lookups.
- Non-Clustered Index on Order(order_date) for reporting.
- Non-Clustered Index on Product(product_name) for search functionality.